NETWORK INTRUSION DETECTION USING MACHINE LEARNING

INTRODUCTION

Modern computer networks are increasingly targeted by cyber attacks such as malware, unauthorized access, Denial-of-Service (DoS), Probe, Remote-to-Local (R2L), and User-to-Root (U2R) attacks. Traditional intrusion detection methods struggle to identify evolving threats efficiently. Machine learning provides an intelligent approach for detecting malicious network activities.

PROBLEM STATEMENT

This project aims to develop a Machine Learning-based Network Intrusion Detection System that automatically analyzes network traffic and classifies it as either normal or malicious. The solution improves cybersecurity through faster, more accurate, and automated cyber threat detection.

OBJECTIVES

• Detect malicious network traffic.
• Classify traffic as normal or attack.
• Improve cybersecurity using machine learning.
• Reduce manual monitoring.
• Select the best model using IBM AutoAI.

PROPOSED SOLUTION

IBM watsonx.ai Studio and AutoAI are used to preprocess data, perform feature engineering, optimize hyperparameters, train multiple machine learning models, and deploy the best-performing model for intrusion detection.

TECHNOLOGIES USED

- IBM Cloud
- IBM watsonx.ai Studio
- IBM AutoAI
- Python
- Machine Learning

DATASET

The dataset contains 40 network traffic features that describe network connections and are used to identify normal and malicious traffic.

MODEL AND RESULTS

Best Model: Decision Tree Classifier (Pipeline 4).

Accuracy: 99.9%

AutoAI performed Feature Engineering (FE), Hyperparameter Optimization (HPO-1 & HPO-2), and model optimization.

Future Scope

The solution can be integrated with enterprise networks, enhanced with new attack types, deployed in cloud environments, and improved using advanced AI models for real-time cybersecurity.

Conclusion

The project demonstrates how IBM watsonx.ai Studio and AutoAI can build an accurate and reliable Network Intrusion Detection System to improve network security.